FACTORING A MAP OF COMPLEXES THROUGH A K-FLAT COMPLEX UP TO HOMOTOPY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The homotopy category of complexes of R-modules is triangulated. Derived Categories, Proposition \ref{derived-proposition-homotopy-category-triangulated}, cited at proof lines 14588-14589. Supplies distinguished triangles, the morphism-of-triangles axioms, and the rotation used to build N.
- Existence of a K-flat resolution with flat terms. Lemma \ref{lemma-K-flat-resolution}, cited at proof line 14593. Provides the quasi-isomorphism M to C with M K-flat and with flat terms.
- K-flatness satisfies two out of three in a distinguished triangle. Lemma \ref{lemma-K-flat-two-out-of-three}, cited at proof line 14600. Shows that the middle term N of the new triangle is K-flat.
- Two out of three quasi-isomorphisms in a morphism of distinguished triangles, and the refinement of a distinguished triangle with an explicit direct sum decomposition. Derived Categories, Lemma \ref{derived-lemma-third-isomorphism-triangle} and Lemma \ref{derived-lemma-improve-distinguished-triangle-homotopy}, cited at proof lines 14619 and 14622-14624. Give the quasi-isomorphism N to L and the improvement of N to a complex with terms M^n direct sum K^n, which preserves flatness.

Source reference. Original lemma 0G6M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20202–20202. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 014S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1483–1483. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8873–8873. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8871–8871. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 014A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1465–1465. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0G6C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20192–20192. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $a : K^\bullet \to L^\bullet$ be a map of complexes of R-modules. If $K^\bullet$ is K-flat, then there exist a complex $N^\bullet$ and maps of complexes $b : K^\bullet \to N^\bullet$ and $c : N^\bullet \to L^\bullet$ such that*

(1) *$N^\bullet$ is K-flat,*
(2) *c is a quasi-isomorphism,*
(3) *a is homotopic to $c \circ b$.*

If the terms of $K^\bullet$ are flat, then we may choose $N^\bullet$, b, and c such that the same is true for $N^\bullet$.

Proof. We will use that the homotopy category $K(R)$ is a triangulated category, see Derived Categories, Proposition 014S. Choose a distinguished triangle $K^\bullet \to L^\bullet \to C^\bullet \to K^\bullet[1]$. Choose a quasi-isomorphism $M^\bullet \to C^\bullet$ with $M^\bullet$ K-flat with flat terms, see Lemma 06Y4. By the axioms of triangulated categories, we may fit the composition $M^\bullet \to C^\bullet \to K^\bullet[1]$ into a distinguished triangle $K^\bullet \to N^\bullet \to M^\bullet \to K^\bullet[1]$. By Lemma 06Y2 we see that $N^\bullet$ is K-flat. Again using the axioms of triangulated categories, we can choose a map $N^\bullet \to L^\bullet$ fitting into the following morphism of distinguished triangles

$$\begin{array}{ccccccc} K^\bullet & \longrightarrow & N^\bullet & \longrightarrow & M^\bullet & \longrightarrow & K^\bullet[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K^\bullet & \longrightarrow & L^\bullet & \longrightarrow & C^\bullet & \longrightarrow & K^\bullet[1] \end{array}$$

Since two out of three of the arrows are quasi-isomorphisms, so is the third arrow $N^\bullet \to L^\bullet$ by the long exact sequences of cohomology associated to these distinguished triangles (or you can look at the image of this diagram in $D(R)$ and use Derived Categories, Lemma 014A if you like). This finishes the proof of (1), (2), and (3). To prove the final assertion, we may choose $N^\bullet$ such that $N^n \cong M^n \oplus K^n$, see Derived Categories, Lemma 0G6C. Hence we get the desired flatness if the terms of $K^\bullet$ are flat. □